

Tentative Schedule

The following is a *tentative* schedule for the course. This is subject to change, and major changes will be announced in lecture if they happen.

Week of . .	Monday	Tuesday	Wednesday	Thursday	Friday
September 15	Introduction & Syllabus	—	Permutations & Combinations	“Quiz”	Axioms of Probability
September 22	Independence, Conditional Probability, Bayes’ Theorem & Total Probability	—	Discrete Random Variables	Quiz	Binomial Random Variables
September 29	Geometric and Poisson Random Variables	—	Continuous Random Variables	Quiz	Binomial, Poisson & Exponential Random Variables
October 6	Normal Distributions	—	Miscellaneous Random Variables	Quiz	Joint Probability Distribution Functions
October 13	Joint Expectation	—	Conditional PDFs & Conditional Expectations	Exam #1	Moment-Generating Functions
October 20	Inequalities, the Law of Large Numbers & the Central Limit Theorem	—	Fall Institute (No Classes)	Quiz	Likelihood Functions & Maximum Likelihood Estimators
October 27	Bayesian Statistics	—	Markov Chains	Quiz	Bivariate Normal Distribution
November 3	Change of Variables	—	Change of Variables, Continued	Exam #2	Vector Spaces & Change of Bases
November 10	Multivariate Normal Distribution	—	Principal Component Analysis	Quiz	Supervised Learning
November 17	Density Estimation & the EM algorithm	—	Reading Day	Reading Day	—